Cross-Task Transfer with Dr. GRPO on Qwen3.5-2B: SOLVE, JUDGE, RECONSTRUCT and a Mixed-Task Model

Full end-to-end report: setup, methods, run settings, results

ANLP project (Turing cluster runs, 2026-10-06 to 2026-10-08)

8 October 2026

Abstract

We train the post-trained `Qwen/Qwen3.5-2B` with Dr. GRPO (LoRA) on three mathematical tasks built from DeepMath-103K: SOLVE (solve a problem), JUDGE (find the first wrong step of a worked solution) and RECONSTRUCT (recover a question from its answer and clues). We train one model per task and one model on an equal mixture of the three tasks, then evaluate every model on all three 400-question test sets.

Each single-task model gains strongly on its own task: SOLVE +16.6 accuracy points, JUDGE +17.2 error-finding points, RECONSTRUCT +21.4 rubric points. Transfer across tasks is mixed: RECONSTRUCT training costs −10.7 on SOLVE.

The mixed-task model (balanced 10/10/10 batches, lr 2×10^{-5}) improves every task with no negative transfer: SOLVE +17.1, JUDGE error-finding +32.2 (Youden's J +45.4), RECONSTRUCT +18.3. It matches the SOLVE specialist, beats the JUDGE specialist, and comes within 3 points of the RECONSTRUCT specialist.

The report also documents why the first SOLVE recipe failed (−2.2) and how it was fixed: a larger batch, a lower learning rate and a stricter, audited answer checker. All numbers use 95% paired bootstrap intervals.

Contents

1 Headline results **3**

2 Tasks, data and prompts **3**
2.1 Model . . . 3
2.2 Data (FINAL-v3) . . . 4
2.3 Prompts and strict gates . . . 4

3 Scoring and evaluation protocol **4**
3.1 SOLVE answer checker (v3) . . . 4
3.2 Judges . . . 5
3.3 Evaluation protocol . . . 5
3.4 Metrics . . . 5

4 Training method: Dr. GRPO **5**
4.1 Objective . . . 5
4.2 Rewards (`rewards.py`) . . . 6

5 Run settings **6**

6 Results in full **7**
6.1 Absolute scores . . . 7
6.2 Δ against the untrained model, all metrics . . . 7
6.3 Mixed model against each specialist . . . 7
6.4 JUDGE by answer-key label . . . 8
6.5 Sub-dimensions (27B judge, 0–4, gated answers count as 0) . . . 8
6.6 Behaviour: gates and answer length (test, all samples) . . . 8
6.7 SOLVE answer types . . . 9

7 Training dynamics **9**

8 The SOLVE story: why the first recipe failed and what fixed it **10**

9 Comparison with published recipes **10**

10 Limitations **11**

11 Compute **11**

12 Reproducibility **11**

A Exact run configurations (`config.json` of each run) **12**
A.1 SOLVE (new recipe), chunk resumed at step 9 12
A.2 Mixed run 12
A.3 JUDGE 13
A.4 RECONSTRUCT 14
A.5 SOLVE (first recipe) 14

B References **15**

1 Headline results

Table 1: Transfer matrix on the test sets: Δ against the untrained Qwen3.5-2B (points, paired, 95% bootstrap CI). Bold marks the model's own training task. JUDGE step % is error-finding (an incorrect solution judged incorrect *and* the exact first wrong step named); Youden's J is verdict accuracy on incorrect plus on correct solutions minus 100.

Trained on ↓ / measured on →	SOLVE accuracy	JUDGE step %	JUDGE Youden J	RECONSTRUCT PC-v4
SOLVE only (step 66)	**+16.6** [+13.8, +19.1]	+8.8 [+2.1, +16.1]	+15.4 [+6.2, +23.8]	+3.8 [+1.4, +6.3]
JUDGE only (step 26)	−1.1 [−4.1, +1.9]	**+17.2** [+10.2, +24.0]	**+28.4** [+19.5, +36.9]	+6.0 [+3.3, +8.6]
RECONSTRUCT only (step 104)	−10.7 [−13.7, −7.7]	+14.2 [+7.2, +21.8]	+7.9 [−1.2, +17.0]	**+21.4** [+19.1, +23.5]
Mixed, 1/3 each (step 75)	**+17.1** [+14.3, +19.9]	**+32.2** [+25.9, +38.6]	**+45.4** [+37.4, +52.6]	**+18.3** [+16.1, +20.5]

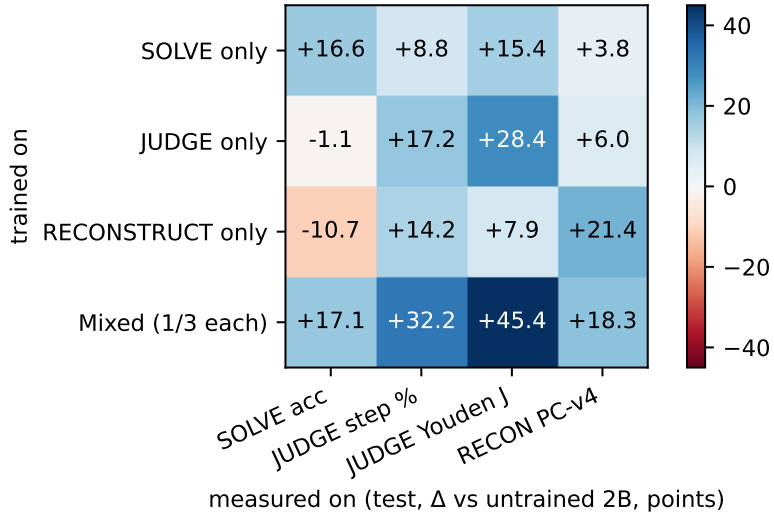


Figure 1: The transfer matrix of Table 1 as a heatmap.

Main findings.

1. **The mixed-task model is the best overall:**
   - on SOLVE, equal to the SOLVE specialist (+0.6, not significant);
   - on JUDGE, far better than the JUDGE specialist (error-finding +15.0, J +17.0);
   - on RECONSTRUCT, 3 points below the RECONSTRUCT specialist (−3.0) but better on same-question recovery (+8.0) and on mathematical validity;
   - every cross-task cell is positive.
2. **Every single-task model improves strongly on its own task.**
3. **Single-task transfer is uneven:**
   - SOLVE helps both other tasks;
   - JUDGE helps RECONSTRUCT and leaves SOLVE unchanged;
   - RECONSTRUCT hurts SOLVE (−10.7), and its JUDGE error-finding gain comes with more false alarms (+10.2).
4. **SOLVE needed a new recipe.** With lr 5×10^{-5} and 8 questions per step it gave −2.2; with lr 10^{-5} and 32 questions per step it gave +16.6 (Section 8).

2 Tasks, data and prompts

2.1 Model

- **Student:** `Qwen/Qwen3.5-2B`, post-trained, non-thinking chat template. Every run starts from it.
- **Reference:** `Qwen/Qwen3.5-9B`, evaluated identically and never trained.
- **Final test judge:** `Qwen/Qwen3.8-27B`, used only for judging (Section 3).

2.2 Data (FINAL-v3)

- **Source:** all questions come from DeepMath-103K (difficulty 5–9).
- **Split per task:** 4,000 rows each, 3,600 train and 400 test.
- **Disjoint and matched:** the three tasks use different questions, matched on topic and difficulty.
- **Dev slice:** a seeded 200-question slice of each train split is never trained on; it is used for checkpoint checks. That leaves 3,400 training questions per task.
- **Training order:** every run uses the same seeded shuffle of those 3,400 (the "matrix order").
- **Leakage check:** no dev or test question appears in any run's training prompts.

Table 2: The three tasks.

Task	Input	Output	How correctness is decided
SOLVE	a maths problem	a worked solution with one `<final_answer>` tag	the answer equals the key (checker v3, deterministic)
JUDGE	a problem and a worked solution split into steps S1…Sn (correct, or with one first error)	`Decision: correct/incorrect`, `First wrong step: Sk/none`, an explanation	verdict and step against the key (parser); explanation by an LLM judge
RECONSTRUCT	a final answer and clues (observations) about a hidden question	one JSON object with `question`, `assumptions`, `justification`	fidelity to the hidden question and validity, by an LLM judge

2.3 Prompts and strict gates

Prompts. The student prompts are version v2 (`baseline.PROMPT_FILES`). Each asks for one final block, carries a brevity line, and states the gate rules.

Strict gates (`strict_gates.py`) run before any score or reward; a failed gate scores 0.

- **SOLVE:** several `<final_answer>` tags that disagree.
- **JUDGE:** several or contradictory decisions or steps, or a decision/step mismatch.
- **RECONSTRUCT:** the answer written into the question, monologue or drafting, or no question. The answer is parsed from the last JSON object.

3 Scoring and evaluation protocol

3.1 SOLVE answer checker (v3)

`solution_verification.compare_answer_detailed` is used for the training reward, every evaluation and the gates. It is deterministic and uses no LLM. Its layers run in order; layers 2–5 can only turn a non-correct decision into "correct":

1. **Rules:**
   - normalised exact match, booleans (with *true/false = yes/no*), letter choices;
   - exact rational arithmetic, congruences, intervals;
   - SymPy equivalence;
   - notation fixes: $\geq$ variants, `\bmod`, `\ln 2`, $+\infty$, degrees, `\sqrt[n]{}`, $|x|$.
2. **Cleaned answer forms:**
   - a variable assignment such as $n = 2^k$ or $(p, q) = (2, 2)$, or $x = 0, y = 0, z = 0$ read as a tuple;
   - a trailing qualifier such as "for $k \geq 1$" removed.

   A cleaned form must be pure mathematics, and a removed part may not contain hedge words (*or, not, maybe, approximately*, …).
3. **Equation chains:** the last side counts as the answer.
4. **Numeric closeness:**
   - exact numbers must be equal exactly;
   - a decimal key with ≥ 4 decimals fixes the precision;
   - infinities are never compared numerically.
5. **DeepMath's official checker pair:** OpenMathInstruct's `math_equal` (vendored; tolerance tightened from 10^{-4} to 10^{-9}; Python `eval` paths removed), then `math-verify`. Both only see pure-mathematics answers that list no more values than the key.
6. **Time limit:** 10 s per check. The main thread uses an alarm; checks from trainer threads run in a process pool.

Audit on 7,465 distinct (key, answer) pairs from training rollouts and test outputs:

- 161 unverified and 8 wrong decisions became correct. Each was inspected by hand.
- 1 correct decision was lost (a hedged answer).
- The false positives found during development are now regression tests:
  - 88327 accepted for 88329 (DeepMath's 10^{-4} tolerance);
  - `True` accepted for 1;
  - $a \neq 3$ read as 3;
  - "3 or 5" accepted for 3;
  - prose such as "does not converge to 0" accepted for 0;
  - $2^{2^{\aleph_0}}$ accepted for $2^{\aleph_0}$ (`math-verify`).

The untrained model moves from 50.2 to 52.4 (dev) and from 51.0 to 52.6 (test). **Every SOLVE number in this report uses v3**; older evaluations were re-scored.

3.2 Judges

- **Final test scores for JUDGE (V/L/E/M) and RECONSTRUCT (PC-v4, same question):** `Qwen3.8-27B` as a logit-reading judge (`qwen_judging.py`): one generated token at temperature 0, probability-weighted levels, calibrated cutoffs. Its agreement with DeepSeek on total scores is $r = 0.940$ (JUDGE) and 0.840 (RECONSTRUCT).
- **Training rewards and RECONSTRUCT dev scoring:** the Jev API (`jev-1.13.0`, `jev_judging.py`): gate → few-shot shortcut stage → calibrated questions.

The judge that selects or rewards a model is never the one that gives its final test score.

3.3 Evaluation protocol

- **Sampling:** temperature 1.0, top-p 1.0, seed 0; a 16k-token first pass, continued to 32k for answers that hit 16k.
- **Samples per question:** SOLVE 4. JUDGE and RECONSTRUCT 1 for trained models; 4 for the untrained 2B and the 9B, whose mean over samples estimates the same single-sample score.
- **Statistics:** deltas are paired on the same questions; 95% bootstrap over questions, 2,000 resamples.

3.4 Metrics

- **SOLVE accuracy:** a strict (gated) verified answer; pass@1 is the mean over 4 samples. **pass@4:** at least one of 4 correct.
- **JUDGE step % (error-finding)**, on the 200 solutions that contain an error: the verdict is "incorrect" *and* the exact first wrong step is named.
- **JUDGE false alarm %:** a correct solution judged incorrect (lower is better).
- **JUDGE Youden's J:** verdict accuracy on incorrect solutions + on correct solutions -100. A one-verdict model scores 0. On this balanced 200/200 set, $J = 2\times$verdict accuracy -100.
- **JUDGE V/L/E/M (/100):** the 27B rubric for verdict, location, explanation and mathematical soundness.
- **RECONSTRUCT PC-v4 (/100):** the 27B rubric for premise fidelity (P), target fidelity (T), evidence use (E) and mathematical validity (M). Shortcuts and gated answers score 0.
- **RECONSTRUCT same question %:** judged equivalent to the hidden original.

Combined JUDGE accuracy (verdict and step, with "none" counted as the step on correct solutions) rewards a bias toward answering "correct". It is reported only next to the per-label split.

4 Training method: Dr. GRPO

4.1 Objective

For each question, $G = 8$ answers $o_1, \ldots, o_G$ are sampled and each gets a reward $r_i \in [0, 1]$.

$$\hat{A}_i = r_i - \tfrac{1}{G}\textstyle\sum_j r_j \qquad \text{(no division by the standard deviation)} \tag{1}$$

$$\mathcal{L} = -\frac{1}{N_{\text{kept}}} \sum_i \sum_t \frac{\rho_{it}\, \hat{A}_i \, \log \pi_\theta(o_{it} \mid q, o_{i,<t})}{16384}, \qquad \rho_{it} = \min\Big(2,\ e^{\log \pi_\theta - \log \pi_{\text{vLLM}}}\Big) \tag{2}$$

- **Normalizer:** the constant 16,384 (the generation cap), not each answer's length (Dr. GRPO).
- **Updates:** one optimizer update per batch, so there is no PPO clipping. KL coefficient $\beta = 0$.
- **Dropped questions:** those whose answers all got the same reward, since they have no signal.
- **Truncated answers** stay in their group with reward 0.

- **Overlapped rollouts** (`async_rollouts`): the answers for step $s{+}1$ are generated with the adapter of step s while step s trains. That is a one-update lag, corrected by ρ; it cut SOLVE's step time from 15 to ≈8 minutes.

4.2 Rewards (`rewards.py`)

- **SOLVE:** $r = 1$ if the single final answer is verified correct (checker v3), else 0.
- **JUDGE:** $r = 0$ for a wrong verdict; otherwise $r = 0.5 + 0.25 \cdot [\text{first wrong step right}] + 0.25 \cdot \min(8, E+M)/8$. Here E (explanation) and M (soundness) are 0–4 Jev levels, probability-weighted.
- **RECONSTRUCT:** $r = 6.25\,(P + T + E + M)/100 - 0.1 \cdot [\text{several drafts}]$, floored at 0, with P, T, E, M from Jev (0–4 each). A shortcut (answer copied, copy-only, no usable question) or a gated answer scores 0.
- **Every task:** a gated or truncated answer scores 0, and a failed Jev call drops the answer from the step.

5 Run settings

Table 3: Settings of every reported run. All runs: LoRA $r = 64$, $\alpha = 64$, dropout 0 on q/k/v/o, in_proj (qkv, z), out_proj, gate/up/down projections (62.5M trainable parameters); AdamW $\beta = (0.9, 0.999)$, weight decay 0, grad clip 1.0; $T = 1.0$, top-p 1.0; 16,384-token cap; Dr. GRPO normalizer 16,384; IS cap 2; $\beta_{\text{KL}} = 0$; truncated answers kept with reward 0, except the RECONSTRUCT run, which filtered them but had 0% truncation; one L40S for vLLM and one for the trainer.

	SOLVE (new)	Mixed	JUDGE	RECONSTRUCT	SOLVE (first)
Run folder (`v3/...`)	`solve-p1-lr1e-5`	`mixed-lr2e-5`	`v3-drgrpo-judge-r2`	`v3-drgrpo-reconstruct`	`v3-drgrpo-solve-r2`
Training data	SOLVE, 3,400	1,134 per task, interleaved S,J,R (3,402)	JUDGE, 3,400	RECONSTRUCT, 3,400	SOLVE, 3,400
Questions × answers / step	32 × 8	30 × 8 (10/10/10)	8 × 8	8 × 8	8 × 8
Learning rate	1×10^{-5}, 5-step warm-up	2×10^{-5}, 5-step warm-up	5×10^{-5} const.	5×10^{-5} const.	5×10^{-5} const.
Rollouts	overlapped (from step 9)	overlapped	synchronous	synchronous	synchronous
Steps trained	79 (stopped by hand)	80 (stopped by hand)	104	104	104
Evaluated checkpoint	66	75	26	104	104
Questions seen by it	2,112	2,250 (750/task)	208	832	832
Selection rule	best 8-step rolling reward, checked on dev	best combined rolling reward (no dev check)	best dev verdict+step of 26/52/78/104	best dev Jev score of 26/52/78/104	best dev accuracy of 26/52/78/104
Wall time per step	≈8 min	≈6.5 min	≈3.6 min	≈1.9 min	≈5 min
GPU-hours (incl. evals)	≈34	≈22.5	≈18	≈10	≈19

Checkpoint tracking (new runs).
- Adapters are kept every 8 steps, plus the latest one with optimizer state.
- The top 3 by rolling training reward are copied to `adapters/best-step`k.
- **Single task:** the score is the 8-step rolling mean reward. In the first pass over the data every step draws unseen questions, so it measures accuracy on new questions.
- **Mixed:** the score is the *combined* rolling reward, the mean over tasks of each task's 8-step rolling reward minus its steps 1–8 average, so that no task dominates by reward scale.
- **Early stopping** (mixed run): after 3 consecutive 8-step checks without beating the best by 0.005. It did not fire, because the user stopped the run at step 80.

Batch composition for the mixed run. Balanced mixed batches were chosen over rotating one-task batches, for four reasons:
1. It is the standard in multi-domain RLVR (GURU, arXiv 2506.14965; Li et al., arXiv 2507.17512).
2. Alternating batches are sequential RL at a one-step period, which shows "mode interference" (arXiv 2602.02301).
3. RL updates for different tasks interfere little (arXiv 2608.03573).
4. GRPO's per-question baselines and the constant Dr. GRPO normalizer give each task a fixed, equal weight per step.

6 Results in full

6.1 Absolute scores

Table 4: Test scores (pass@1 unless noted; 95% CI). False alarm: lower is better.

Model	SOLVE		JUDGE					RECONSTRUCT	
	acc	pass@4	verdict	step %	false al.	J	V/L/E/M	PC-v4	same Q
Untrained 2B	52.6	80.2	64.3	34.8	37.2	28.6	53.9	40.1	2.9
SOLVE only (66)	69.2	**87.5**	72.0	43.5	23.5	44.0	62.9	43.9	6.8
JUDGE only (26)	51.6	76.0	78.5	52.0	13.0	57.0	72.2	46.1	5.8
RECONSTRUCT only (104)	41.9	66.2	68.2	49.0	47.5	36.5	55.7	**61.4**	2.5
Mixed (75)	**69.8**	84.5	**87.0**	**67.0**	**7.5**	**74.0**	**80.8**	58.4	**10.5**
Qwen3.5-9B (reference)	82.7	91.5	87.6	87.9	17.8	75.2	85.7	72.7	19.6

95% CIs for the rows of Table 4:

- SOLVE accuracy: untrained [48.9, 56.2]; SOLVE model [65.6, 72.9]; JUDGE model [48.0, 55.1]; RECONSTRUCT model [38.4, 45.6]; mixed [66.1, 73.4].
- JUDGE step %: untrained [30.9, 38.5]; SOLVE model [36.5, 50.5]; JUDGE model [45.0, 58.5]; RECONSTRUCT model [42.5, 56.0]; mixed [60.0, 73.5].
- RECONSTRUCT PC-v4: untrained [38.4, 41.9]; SOLVE model [41.2, 46.6]; JUDGE model [43.4, 48.8]; RECONSTRUCT model [59.3, 63.3]; mixed [55.9, 60.8].

6.2 Δ against the untrained model, all metrics

Table 5: Δ vs untrained 2B (points; paired 95% CI).

Trained on	SOLVE acc	pass@4	verdict	step %	false al.	J	V/L/E/M	PC-v4	same Q
SOLVE (66)	+16.6 [13.8, 19.1]	+7.2 [3.8, 10.8]	+7.7 [3.1, 11.9]	+8.8 [2.1, 16.1]	−13.8 [−19.8, −8.1]	+15.4 [6.2, 23.8]	+9.0 [5.1, 12.6]	+3.8 [1.4, 6.3]	+3.9 [1.9, 5.9]
JUDGE (26)	−1.1 [−4.1, 1.9]	−4.2 [−8.0, −0.2]	+14.2 [9.8, 18.4]	+17.2 [10.2, 24.0]	−24.2 [−29.2, −19.4]	+28.4 [19.5, 36.9]	+18.3 [14.7, 21.8]	+6.0 [3.3, 8.6]	+2.9 [0.8, 4.9]
RECON (104)	−10.7 [−13.7, −7.7]	−14.0 [−18.2, −9.8]	+3.9 [−0.6, 8.5]	+14.2 [7.2, 21.8]	+10.2 [3.9, 16.5]	+7.9 [−1.2, 17.0]	+1.8 [−2.0, 5.6]	+21.4 [19.1, 23.5]	−0.4 [−2.2, 1.4]
Mixed (75)	+17.1 [14.3, 19.9]	+4.2 [1.0, 7.8]	+22.7 [18.7, 26.3]	+32.2 [25.9, 38.6]	−29.8 [−34.2, −25.5]	+45.4 [37.4, 52.6]	+26.9 [23.7, 29.9]	+18.3 [16.1, 20.5]	+7.6 [4.9, 10.3]

6.3 Mixed model against each specialist

Table 6: Paired Δ = mixed − specialist (95% CI). Bold: the specialist's own task.

Compared with	SOLVE acc	pass@4	step %	false al.	J	V/L/E/M	PC-v4	same Q
SOLVE model (66)	**+0.6** [−1.8, 2.8]	−3.0 [−5.8, −0.2]	+23.5 [15.0, 32.5]	−16.0 [−22.5, −9.5]	+30.0 [20.5, 40.0]	+17.9 [13.6, 22.2]	+14.5 [11.4, 17.5]	+3.8 [0.8, 6.8]
JUDGE model (26)	+18.2 [15.1, 21.2]	+8.5 [5.0, 12.0]	**+15.0** [7.5, 23.0]	−5.5 [−10.5, −0.5]	**+17.0** [8.0, 25.5]	**+8.5** [5.0, 12.2]	+12.3 [9.4, 15.2]	+4.8 [2.0, 7.8]
RECON model (104)	+27.8 [24.4, 31.4]	+18.2 [13.8, 22.8]	+18.0 [8.5, 27.5]	−40.0 [−47.5, −33.0]	+37.5 [26.5, 48.0]	+25.1 [20.4, 29.5]	−3.0 [−5.9, −0.1]	**+8.0** [5.0, 11.0]

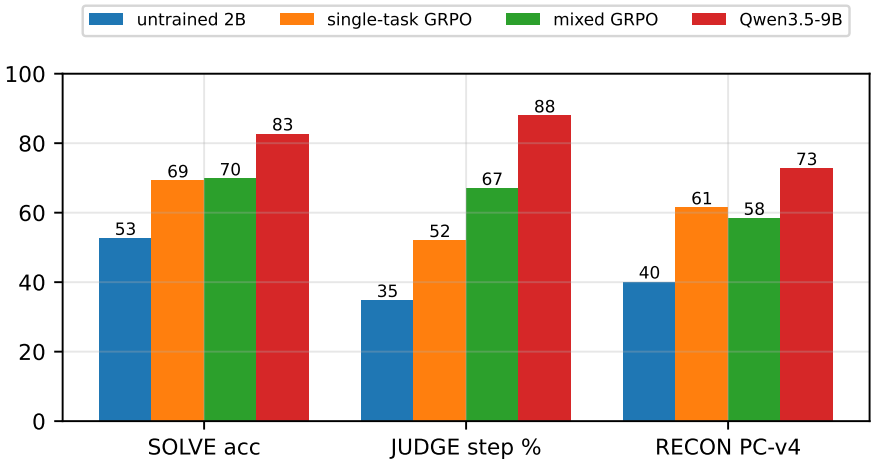


Figure 2: Own-task scores: untrained 2B, single-task GRPO, mixed GRPO, and the untrained 9B.

How much of the gap to the 9B is closed. Gap closed = (model − 2B)/(9B − 2B):

Metric	Single-task specialist	Mixed model
SOLVE accuracy	55%	57%
JUDGE error-finding	32%	61%
RECONSTRUCT PC-v4	66%	56%

On JUDGE the mixed model nearly matches the 9B: verdict accuracy 87.0 vs 87.6, and fewer false alarms (7.5 vs 17.8).

6.4 JUDGE by answer-key label

Table 7: JUDGE verdict accuracy by key label, and error-finding (200 questions each).

Model	correct solutions: verdict	incorrect solutions: verdict	incorrect: verdict + step
Untrained 2B	62.7	65.9	34.8
SOLVE only	76.5	67.5	43.5
JUDGE only	87.0	70.0	52.0
RECONSTRUCT only	52.5	84.0	49.0
Mixed	**92.5**	81.5	**67.0**

Reading the table.
- The RECONSTRUCT model's gain on incorrect solutions comes from answering "incorrect" more often; its accuracy on correct solutions falls to 52.5.
- The mixed model improves on both kinds of solution.

6.5 Sub-dimensions (27B judge, 0–4, gated answers count as 0)

Model	JUDGE V	L	E	M	RECON P	T	E	M
Untrained 2B	2.57	1.95	2.08	1.99	1.71	1.97	1.90	0.84
SOLVE only	2.88	2.40	2.52	2.26	1.88	2.10	2.06	0.98
JUDGE only	3.14	2.78	2.80	2.83	1.88	2.17	2.28	1.04
RECONSTRUCT only	2.73	2.03	2.09	2.06	**2.49**	**3.00**	**3.58**	0.75
Mixed	**3.48**	**3.19**	**3.25**	**3.00**	2.37	2.72	2.88	**1.37**
Qwen3.5-9B	3.61	3.50	3.54	3.50	2.85	3.17	3.44	2.33

- The RECONSTRUCT specialist gains on fidelity and evidence use (P, T, E), but its mathematical validity drops (0.84 → 0.75).
- The mixed model has the best validity of the 2B models (1.37).

6.6 Behaviour: gates and answer length (test, all samples)

Model	SOLVE		JUDGE		RECONSTRUCT	
	gate %	mean tokens	gate %	mean tokens	gate %	mean tokens
Untrained 2B	6.8	8,089	13.8	5,087	12.6	5,337
SOLVE only	1.4	8,643	11.8	5,613	11.5	5,734
JUDGE only	4.0	5,518	1.2	3,017	16.0	5,287
RECONSTRUCT only	1.1	5,291	3.8	5,026	3.8	2,236
Mixed	1.0	9,114	1.8	6,751	5.5	7,793
Qwen3.5-9B	3.2	4,088	5.4	3,737	3.8	3,889

- JUDGE and RECONSTRUCT training shorten answers on their own tasks (−41% and −58%).
- The mixed model writes longer answers on every task (SOLVE +13%, JUDGE +33%, RECONSTRUCT +46%).
- No trained model has more than 0.5% of answers capped at 32k.

6.7 SOLVE answer types

Model	correct	wrong	unverified	gated	no answer	capped
Untrained 2B	52.6	20.8	16.2	6.8	3.6	0.1
SOLVE only (66)	69.2	15.6	12.6	1.4	1.3	0.0
Mixed (75)	69.8	16.4	11.4	1.0	1.0	0.4

About 8 points of the SOLVE gain come from format fixes (gated or missing answers); about 9 come from fewer wrong or undecidable answers.

7 Training dynamics

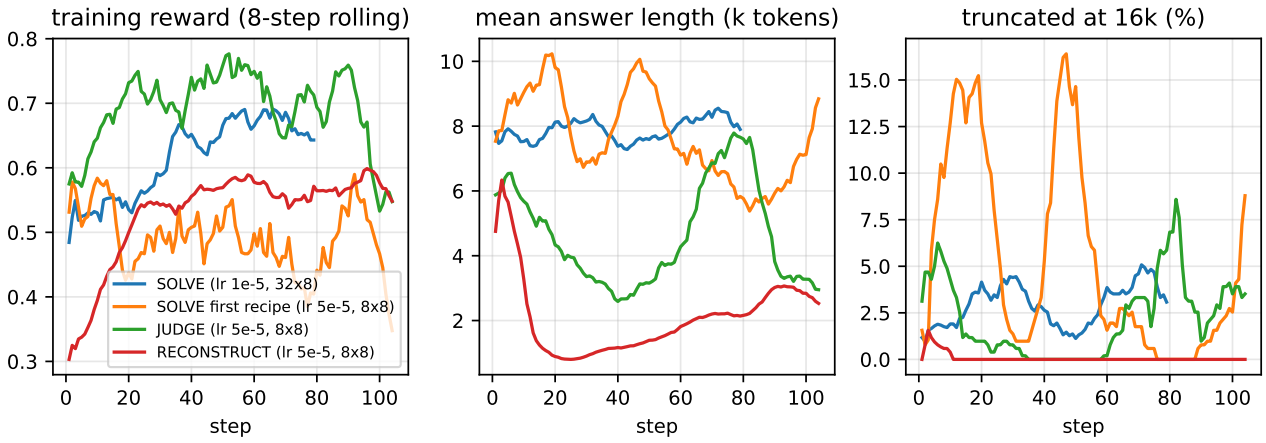


Figure 3: Single-task runs. Rewards differ by task and are comparable only within a curve.

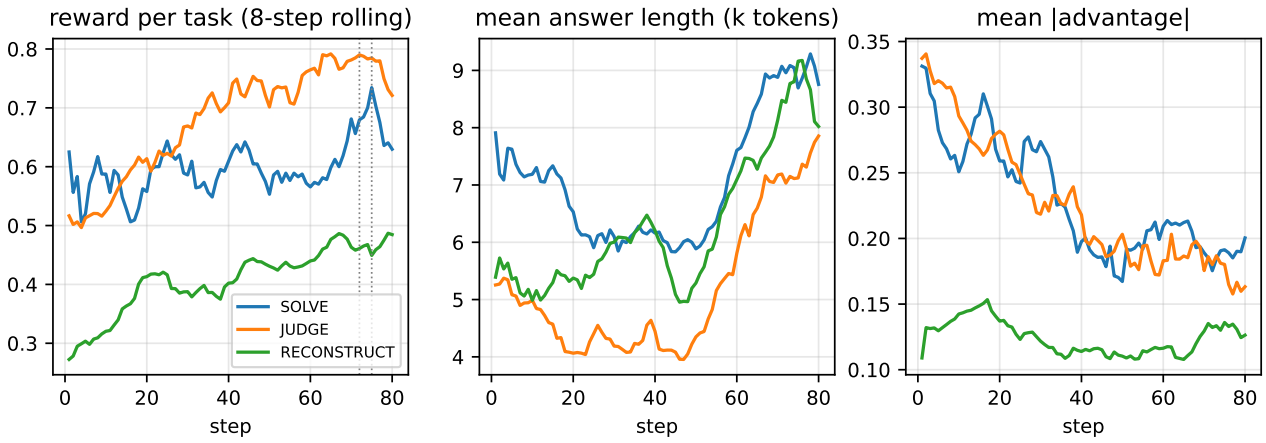


Figure 4: Mixed run, per task. Dotted lines: steps 72 and 75.

Table 8: Mixed run in 8-step blocks.

Steps	SOLVE reward	JUDGE reward	RECON reward	length S / J / R	truncation
1–8	0.62	0.52	0.31	7.1k / 4.9k / 5.1k	1–5%
17–24	0.62	0.62	0.42	6.1k / 4.3k / 5.4k	0–1%
41–48	0.59	0.75	0.44	6.0k / 4.1k / 5.0k	0%
65–72	0.68	0.79	0.46	9.0k / 7.0k / 8.4k	≈4%
73–80	0.63	0.72	0.49	8.8k / 7.9k / 8.0k	4–7%

Combined rolling reward at the 8-step checks:

Step	16	24	32	40	48	56	64	72	80
Combined gain	+0.007	+0.072	+0.065	+0.091	+0.109	+0.092	+0.128	+0.161	+0.130

Its per-step maximum is +0.174, at step 75.

Observations.
- **Reward scale:** RECONSTRUCT's mean $|\hat{A}|$ was about 2× smaller than SOLVE's and JUDGE's (0.11–0.15 vs 0.20–0.34). That is below the planned 3× threshold, so no per-task rescaling was applied.
- **Length drift:** answer lengths rose for every task from step ≈57 at lr 2×10^{-5}. This is the same drift the lr 5×10^{-5} runs showed.
- **JUDGE early peak:** in its single-task run, JUDGE peaked early (dev best at step 26) and declined. The SOLVE and RECONSTRUCT rewards plateaued after steps ≈50 and ≈40.

8 The SOLVE story: why the first recipe failed and what fixed it

First recipe (lr 5×10^{-5}, 8×8, 104 steps): test -2.2 $[-5.8, +1.1]$. Diagnosis from its 6,656 training rollouts:
- **Too little signal per step.** Questions whose answers differed fell from 5.5 to 3.8 of 8, so each update came from about 3 questions at a high lr.
- **More wrong answers.** Wrong answers rose from 23% to 33% of rollouts. Truncation fell, but the formerly truncated answers now finished wrong. The rolling reward fell from 0.55 to 0.35.
- **Little data.** Only 832 of 3,400 questions were seen.

Fixes and evidence.
1. **Checker v3** (Section 3).
2. **32 questions × 8 answers per step**, so 18–25 questions per step carry a signal.
3. **lr probe**, at 32×8, step 8, dev 200×4:
   - 10^{-5}: $+3.8$ $[-0.2, +7.6]$;
   - 2×10^{-5}: $+3.9$ $[+0.2, +7.5]$;
   - the first recipe at step 26 was -4.5.

   We chose 10^{-5}, the literature value for LoRA (≈ 10× the 10^{-6} used for full fine-tuning in DeepMath and DeepScaleR).
4. **Overlapped rollouts** (15 → 8 min/step); the log-prob gap stayed at 0.010–0.011.
5. **Result:** the rolling reward rose from 0.53 to 0.69 (step 66). Dev at step 66 was 72.4 vs 52.4 ($+20.0$ $[+15.9, +24.1]$); test +16.6.

Deliberately not used:
- difficulty filtering (user decision);
- length penalties;
- a longer cap: 16k already covers 99.3% of the untrained model's correct answers.

Dev checkpoint scores of the other runs.
- **JUDGE**, verdict + step (sum of two accuracies) at steps 26/52/78/104: **1.441**, 1.215, 1.302, 1.034.
- **RECONSTRUCT**, Jev score: 56.6, 62.2, 60.7, **63.1**.
- **First SOLVE recipe**, checker v3: 47.9, 51.2, 49.4, **54.2** (untrained 52.4).

9 Comparison with published recipes

Recipe	lr	questions × answers / step	reward	notes
Ours, SOLVE (new)	10^{-5} LoRA	32 × 8	1/0 verified	Dr. GRPO, no KL, overlapped
Ours, mixed	2×10^{-5} LoRA	30 × 8 (10/10/10)	per task	as above
DeepMath-1.5B (official code)	10^{-6} full	128 × 16, refills groups	±1 (math_equal ∨ math-verify)	KL 0.001, T 0.6, 24k
DeepScaleR-1.5B	10^{-6} full	128 × 8	1/0	8k→16k→24k
Dr. GRPO paper	full	–	1/0	no ÷std, constant normalizer
TRL "LoRA without regret"	≈ 10^{-5} LoRA	32–512	–	LoRA ≈ 10× full-FT lr

10 Limitations

1. **One seed per model.** The confidence intervals cover question sampling only.
2. **The models use different recipes:**
   - SOLVE: lr 10^{-5}, 32×8;
   - mixed: lr 2×10^{-5}, 30×8;
   - JUDGE and RECONSTRUCT: lr 5×10^{-5}, 8×8.

   The cross-task and mixed-vs-specialist comparisons therefore mix task and recipe effects. In particular, the JUDGE specialist saw 208 JUDGE questions, against 750 for the mixed model.
3. **Different selection rules.** The mixed model's checkpoint was chosen by training reward without a dev check.
4. **One sample** per question for trained models on JUDGE and RECONSTRUCT.
5. **Judge dependence:** the judge-based metrics depend on the 27B judge's calibration (r 0.94 / 0.84 vs DeepSeek).
6. **Unverifiable answers:** about 11–16% of SOLVE answers stay unverified (forms that rules cannot check) and count as wrong for every model. Deltas are unaffected; absolute accuracy is underestimated.
7. **Longer answers:** the mixed model writes longer answers on every task.

11 Compute

- **Hardware:** the Turing cluster, NVIDIA L40S (46 GB).
- **GPU limits:** at most 6 GPUs at once, then at most 4 from 2026-10-07.
- **Ledger:** 234 GPU-hours used of the 500 approved, covering all work (baselines, judge calibration, the first matrix, the SOLVE probes and run, the mixed run, all evaluations and judging). 266 remain.
- **Jev API:** JUDGE and RECONSTRUCT rewards and RECONSTRUCT dev scoring, a few US dollars in total.

12 Reproducibility

```
# SOLVE model (lr 1e-5, 32x8), end to end from the workstation
nohup bash scripts/solve_grpo_pipeline.sh \
    > artifacts/opd/evals/solve-pipeline.log 2>&1 &
# Mixed model (1/3 each, lr 2e-5, 30x8), end to end
NODE=node07 nohup bash scripts/mixed_grpo_pipeline.sh \
    > artifacts/opd/evals/mixed-pipeline.log 2>&1 &

# one run by hand (2 GPUs: vLLM + trainer)
python3 scripts/turing/launch.py submit --name <run> --gpus 2 --hours 14 --node <node> -- \
  python scripts/turing/opd_run.py --task {solve|judge|reconstruct|mixed} --config grpo \
    --steps <n> --set prompts_per_step=<32|30> --set optimizer.lr=<1e-5|2e-5> \
    --set optimizer.warmup_steps=5 --set async_rollouts=true \
    --set best_window=8 --set best_keep=3 --student-port <port> \
    --out /scratch/mukund.hebbar/anlp-opd/v3/<run>

# evaluation, 27B judging, calibration, reports
python scripts/turing/eval_matrix.py --name <n> --adapter <adapter> --split test \
    --samples 4 --samples-tasks solve --out <dir>
python scripts/turing/qwen_judge_run.py --eval-dir test:<dir>/s0 --out <log>/answers.jsonl
uv run python scripts/qwen_judge_calibration.py score <dir>/s0
uv run python scripts/rescore_solve.py <eval dirs>          # after any checker change
SOLVE_SEL=66 MIXED_SEL=75 uv run python scripts/report_grpo_matrix.py
uv run python scripts/make_report_figures.py
uv run python -m unittest discover -s tests                 # 229 tests
```

Files.

- **Test answers and scores:** `artifacts/baselines/final-v3/test/`, in the folders
  - `v3-pass4-2b` (untrained 2B) and `v3-pass4-9b` (9B);
  - `v3-grpo-lr1e-5-solve-step66` (SOLVE);
  - `v3-drgrpo-judge-step26` (JUDGE) and `v3-drgrpo-reconstruct-step104` (RECONSTRUCT);
  - `v3-grpo-mixed-lr2e-5-step75` (mixed);

  - v3-drgrpo-solve-step104 (first SOLVE recipe).
- **Training metrics and checkpoint records:** artifacts/opd/evals/{solve-p1-lr1e-5, mixed-lr2e-5, v3-drgrpo-dev}.
- **Adapters** (Turing scratch):
  - SOLVE: node11 v3/solve-p1-lr1e-5/adapters/best-step66, also copied to ~/anlp-project/runs-out;
  - mixed: node07 v3/mixed-lr2e-5/adapters/best-step75;
  - JUDGE: node11 v3/v3-drgrpo-judge-r2/adapters/step26;
  - RECONSTRUCT: node07 v3/v3-drgrpo-reconstruct/adapters/step104.
- **Code:**
  - src/anlp_project/: opd_train.py (GRPO loop, overlap, per-task metrics, best checkpoints), opd.py (configs, mixed prompts), rewards.py, solution_verification.py (checker v3), strict_gates.py, jev_judging.py, qwen_judging.py;
  - scripts in scripts/ and scripts/turing/.

A Exact run configurations (config.json of each run)

Each file is as written by the trainer at the start of its final chunk (one key per line); overrides lists the command-line -set values. The fields teacher, objective, loss_normalization, topk_*, sft_mix, verdict_reward and teacher_* are inherited from the on-policy-distillation configuration and are *unused* when method is grpo. The GRPO loss is the one in Section 4, with norm_tokens as the constant normalizer.

A.1 SOLVE (new recipe), chunk resumed at step 9


```
{
  "task": "solve",
  "steps": 81,
  "platform": "turing-l40s",
  "gpus": ["0", "1"],
  "overrides": ["prompts_per_step=32", "optimizer.lr=1e-5", "optimizer.warmup_steps=5", "async_rollouts=true",
      "best_window=8", "best_keep=3"],
  "prompts": "matrix",
  "config_name": "grpo",
  "external_teacher": false,
  "init_adapter": null,
  "resume_from": "/scratch/mukund.hebbar/anlp-opd/v3/solve-p1-lr1e-5/adapters/step9",
  "start_step": 9,
  "student": "Qwen/Qwen3.5-2B",
  "teacher": null,
  "objective": "per-token reverse KL, sampled-token estimator, discount 0",
  "loss_normalization": "token mean per rollout, then mean over rollouts",
  "importance_sampling_cap": 2.0,
  "lora": {"r": 64, "alpha": 64, "dropout": 0.0, "target_modules": ["q_proj", "k_proj", "v_proj", "o_proj",
      "in_proj_qkv", "in_proj_z", "out_proj", "gate_proj", "up_proj", "down_proj"]},
  "optimizer": {"name": "adamw", "lr": 1e-05, "betas": [0.9, 0.999], "weight_decay": 0.0, "warmup_steps": 5,
      "grad_clip": 1.0},
  "prompts_per_step": 32,
  "samples_per_prompt": 8,
  "updates_per_step": 1,
  "rollout": {"temperature": 1.0, "top_p": 1.0},
  "rollout_max_tokens": {"solve": 16384, "judge": 16384, "reconstruct": 16384},
  "logprob_chunk_tokens": 1024,
  "auto_stop": {"truncation_rise": 0.15, "truncation_ceiling": 0.35, "patience": 3, "max_mean_abs_logprob_gap": 0.5,
      "repetition_threshold": 0.5, "repetition_patience": 2},
  "drop_truncated": false,
  "kl_anchor_beta": 0.0,
  "overlong_penalty": 0.0,
  "soft_length_penalty": null,
  "teacher_system_prompt": null,
  "loss": "sampled",
  "teacher_signal_max_len": null,
  "sft_mix": null,
  "verdict_reward": null,
  "topk_k": 20,
  "topk_divergence": "reverse",
  "train_micro_tokens": 16384,
  "keep_adapters_every": 8,
  "method": "grpo",
  "variant": "dr_grpo",
  "norm_tokens": 16384,
  "reward_timeout_s": 180,
  "async_rollouts": true,
  "best_window": 8,
  "best_keep": 3,
}
```


A.2 Mixed run


```
{
```

```
  "task": "mixed",
  "steps": 90,
  "platform": "turing-l40s",
  "gpus": ["0", "1"],
  "overrides": ["prompts_per_step=30", "optimizer.lr=2e-5", "optimizer.warmup_steps=5", "async_rollouts=true",
      "best_window=8", "best_keep=3"],
  "prompts": "matrix",
  "config_name": "grpo",
  "external_teacher": false,
  "init_adapter": null,
  "resume_from": null,
  "start_step": 0,
  "student": "Qwen/Qwen3.5-2B",
  "teacher": null,
  "objective": "per-token reverse KL, sampled-token estimator, discount 0",
  "loss_normalization": "token mean per rollout, then mean over rollouts",
  "importance_sampling_cap": 2.0,
  "lora": {"r": 64, "alpha": 64, "dropout": 0.0, "target_modules": ["q_proj", "k_proj", "v_proj", "o_proj",
      "in_proj_qkv", "in_proj_z", "out_proj", "gate_proj", "up_proj", "down_proj"]},
  "optimizer": {"name": "adamw", "lr": 2e-05, "betas": [0.9, 0.999], "weight_decay": 0.0, "warmup_steps": 5,
      "grad_clip": 1.0},
  "prompts_per_step": 30,
  "samples_per_prompt": 8,
  "updates_per_step": 1,
  "rollout": {"temperature": 1.0, "top_p": 1.0},
  "rollout_max_tokens": {"solve": 16384, "judge": 16384, "reconstruct": 16384},
  "logprob_chunk_tokens": 1024,
  "auto_stop": {"truncation_rise": 0.15, "truncation_ceiling": 0.35, "patience": 3, "max_mean_abs_logprob_gap": 0.5,
      "repetition_threshold": 0.5, "repetition_patience": 2},
  "drop_truncated": false,
  "kl_anchor_beta": 0.0,
  "overlong_penalty": 0.0,
  "soft_length_penalty": null,
  "teacher_system_prompt": null,
  "loss": "sampled",
  "teacher_signal_max_len": null,
  "sft_mix": null,
  "verdict_reward": null,
  "topk_k": 20,
  "topk_divergence": "reverse",
  "train_micro_tokens": 16384,
  "keep_adapters_every": 8,
  "method": "grpo",
  "variant": "dr_grpo",
  "norm_tokens": 16384,
  "reward_timeout_s": 180,
  "async_rollouts": true,
  "best_window": 8,
  "best_keep": 3,
}
```


A.3 JUDGE


```
{
  "task": "judge",
  "steps": 104,
  "platform": "turing-l40s",
  "gpus": ["0", "1"],
  "overrides": ["drop_truncated=false"],
  "prompts": "matrix",
  "config_name": "grpo",
  "external_teacher": false,
  "init_adapter": null,
  "resume_from": null,
  "start_step": 0,
  "student": "Qwen/Qwen3.5-2B",
  "teacher": null,
  "objective": "per-token reverse KL, sampled-token estimator, discount 0",
  "loss_normalization": "token mean per rollout, then mean over rollouts",
  "importance_sampling_cap": 2.0,
  "lora": {"r": 64, "alpha": 64, "dropout": 0.0, "target_modules": ["q_proj", "k_proj", "v_proj", "o_proj",
      "in_proj_qkv", "in_proj_z", "out_proj", "gate_proj", "up_proj", "down_proj"]},
  "optimizer": {"name": "adamw", "lr": 5e-05, "betas": [0.9, 0.999], "weight_decay": 0.0, "warmup_steps": 0,
      "grad_clip": 1.0},
  "prompts_per_step": 8,
  "samples_per_prompt": 8,
  "updates_per_step": 1,
  "rollout": {"temperature": 1.0, "top_p": 1.0},
  "rollout_max_tokens": {"solve": 16384, "judge": 16384, "reconstruct": 16384},
  "logprob_chunk_tokens": 1024,
  "auto_stop": {"truncation_rise": 0.15, "truncation_ceiling": 0.35, "patience": 3, "max_mean_abs_logprob_gap": 0.5,
      "repetition_threshold": 0.5, "repetition_patience": 2},
  "drop_truncated": false,
  "kl_anchor_beta": 0.0,
  "overlong_penalty": 0.0,
  "soft_length_penalty": null,
  "teacher_system_prompt": null,
  "loss": "sampled",
  "teacher_signal_max_len": null,
```

```
  "sft_mix": null,
  "verdict_reward": null,
  "topk_k": 20,
  "topk_divergence": "reverse",
  "train_micro_tokens": 16384,
  "keep_adapters_every": 8,
  "method": "grpo",
  "variant": "dr_grpo",
  "norm_tokens": 16384,
  "reward_timeout_s": 180,
}
```


A.4 RECONSTRUCT


```
{
  "task": "reconstruct",
  "steps": 104,
  "platform": "turing-l40s",
  "gpus": ["0", "1"],
  "overrides": [],
  "prompts": "matrix",
  "config_name": "grpo",
  "external_teacher": false,
  "init_adapter": null,
  "resume_from": null,
  "start_step": 0,
  "student": "Qwen/Qwen3.5-2B",
  "teacher": null,
  "objective": "per-token reverse KL, sampled-token estimator, discount 0",
  "loss_normalization": "token mean per rollout, then mean over rollouts",
  "importance_sampling_cap": 2.0,
  "lora": {"r": 64, "alpha": 64, "dropout": 0.0, "target_modules": ["q_proj", "k_proj", "v_proj", "o_proj",
      "in_proj_qkv", "in_proj_z", "out_proj", "gate_proj", "up_proj", "down_proj"]},
  "optimizer": {"name": "adamw", "lr": 5e-05, "betas": [0.9, 0.999], "weight_decay": 0.0, "warmup_steps": 0,
      "grad_clip": 1.0},
  "prompts_per_step": 8,
  "samples_per_prompt": 8,
  "updates_per_step": 1,
  "rollout": {"temperature": 1.0, "top_p": 1.0},
  "rollout_max_tokens": {"solve": 16384, "judge": 16384, "reconstruct": 16384},
  "logprob_chunk_tokens": 1024,
  "auto_stop": {"truncation_rise": 0.15, "truncation_ceiling": 0.35, "patience": 3, "max_mean_abs_logprob_gap": 0.5,
      "repetition_threshold": 0.5, "repetition_patience": 2},
  "drop_truncated": true,
  "kl_anchor_beta": 0.0,
  "overlong_penalty": 0.0,
  "soft_length_penalty": null,
  "teacher_system_prompt": null,
  "loss": "sampled",
  "teacher_signal_max_len": null,
  "sft_mix": null,
  "verdict_reward": null,
  "topk_k": 20,
  "topk_divergence": "reverse",
  "train_micro_tokens": 16384,
  "keep_adapters_every": 8,
  "method": "grpo",
  "variant": "dr_grpo",
  "norm_tokens": 16384,
  "reward_timeout_s": 180,
}
```


A.5 SOLVE (first recipe)


```
{
  "task": "solve",
  "steps": 104,
  "platform": "turing-l40s",
  "gpus": ["0", "1"],
  "overrides": ["drop_truncated=false"],
  "prompts": "matrix",
  "config_name": "grpo",
  "external_teacher": false,
  "init_adapter": null,
  "resume_from": null,
  "start_step": 0,
  "student": "Qwen/Qwen3.5-2B",
  "teacher": null,
  "objective": "per-token reverse KL, sampled-token estimator, discount 0",
  "loss_normalization": "token mean per rollout, then mean over rollouts",
  "importance_sampling_cap": 2.0,
  "lora": {"r": 64, "alpha": 64, "dropout": 0.0, "target_modules": ["q_proj", "k_proj", "v_proj", "o_proj",
      "in_proj_qkv", "in_proj_z", "out_proj", "gate_proj", "up_proj", "down_proj"]},
  "optimizer": {"name": "adamw", "lr": 5e-05, "betas": [0.9, 0.999], "weight_decay": 0.0, "warmup_steps": 0,
      "grad_clip": 1.0},
  "prompts_per_step": 8,
  "samples_per_prompt": 8,
  "updates_per_step": 1,
```

```
  "rollout": {"temperature": 1.0, "top_p": 1.0},
  "rollout_max_tokens": {"solve": 16384, "judge": 16384, "reconstruct": 16384},
  "logprob_chunk_tokens": 1024,
  "auto_stop": {"truncation_rise": 0.15, "truncation_ceiling": 0.35, "patience": 3, "max_mean_abs_logprob_gap": 0.5,
      "repetition_threshold": 0.5, "repetition_patience": 2},
  "drop_truncated": false,
  "kl_anchor_beta": 0.0,
  "overlong_penalty": 0.0,
  "soft_length_penalty": null,
  "teacher_system_prompt": null,
  "loss": "sampled",
  "teacher_signal_max_len": null,
  "sft_mix": null,
  "verdict_reward": null,
  "topk_k": 20,
  "topk_divergence": "reverse",
  "train_micro_tokens": 16384,
  "keep_adapters_every": 8,
  "method": "grpo",
  "variant": "dr_grpo",
  "norm_tokens": 16384,
  "reward_timeout_s": 180,
}
```


B References

- Liu et al., *Understanding R1-Zero-Like Training: A Critical Perspective* (Dr. GRPO), arXiv 2503.20783.
- He et al., *DeepMath-103K*, arXiv 2504.11456; official code `https://github.com/zwhe99/DeepMath`.
- Luo et al., *DeepScaleR: Surpassing O1-Preview with a 1.5B Model by Scaling RL* (2025).
- Cheng et al., *Revisiting RL for LLM Reasoning from a Cross-Domain Perspective* (GURU), arXiv 2506.14965.
- Li et al., *Can One Domain Help Others? A Data-Centric Study on Multi-Domain Reasoning via RL*, arXiv 2507.17512.
- *Advancing General-Purpose Reasoning Models with Modular Gradient Surgery*, arXiv 2602.02301.
- *SFT Conflicts, RL Coexists: Multi-Task Learning for LLMs*, arXiv 2608.03573.
- Hugging Face TRL, *LoRA Without Regret*; `math-verify`; OpenMathInstruct grader (via DeepMath).